UAV-BASED VEHICLE DETECTION AND TRAJECTORY ANALYSIS FOR IMPROVING HIGHWAY TRAFFIC SAFETY

INTERIM PROJECT REPORT

Submitted by

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To

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of
Master of Technology
in
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DEPARTMENT OF CIVIL ENGINEERING

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DECLARATION

I hereby declare that the interim project report titled "UAV-Based Vehicle Detection and Trajectory Analysis for Improving Highway Traffic Safety", submitted in partial fulfilment of the requirements for the award of the degree of Master of Technology at APJ Abdul Kalam Technological University, Kerala, is the original work carried out by me under the supervision of Prof. Aparna V.

I confirm that this report is the result of my independent efforts and that all sources of information used have been appropriately acknowledged. I have maintained academic honesty and integrity throughout the preparation of this report and have neither fabricated data nor presented others' work as my own without proper citation.

I understand that any breach of academic ethics, such as plagiarism or data falsification, may lead to disciplinary proceedings by the institute and/or the University. I also affirm that this report has not been submitted previously for the award of any degree, diploma, or other qualification at any other institution.

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CERTIFICATE

This is to certify that this interim project report entitled ***"UAV-Based Vehicle Detection and Trajectory Analysis for Improving Highway Traffic Safety"*** *is a bonafide record of the interim project work presented by* ***ERIC JOSE (University Reg. No: TVE25CEGI05)*** *to the APJ Abdul Kalam Technological University for the award of the Degree of Master of Technology in Geoinformatics, Civil Engineering, during the academic year 2026-2027, carried out under my guidance and supervision. This report has not been submitted to any other university or institute for any purpose.*

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ABSTRACT

Road traffic safety assessment increasingly needs vehicle-level information on how vehicles move and interact. Conventional crash records, manual surveys and fixed sensors are retrospective, costly or limited in coverage. Unmanned Aerial Vehicles (UAVs) provide a flexible bird's-eye view from which vehicle trajectories can be extracted over a road section. This study proposes an **improved YOLO-based vehicle detection and trajectory analysis framework** for traffic safety monitoring, crash risk assessment and accident hotspot identification on mid-block highway sections in Thiruvananthapuram.

A review of 17 studies shows that UAV trajectories can support time-to-collision (TTC), post-encroachment time (PET) and extreme-value-based crash risk estimation, but that small vehicles, camera motion, mixed traffic and georeferencing remain open problems, especially for Indian traffic with five vehicle classes: two-wheelers, three-wheelers, cars, light commercial vehicles and heavy commercial vehicles. The methodology covers UAV flight planning with DGPS ground control, CVAT annotation, baseline and small-object-optimised YOLO detectors, multi-object tracking, DGPS georeferencing, vehicle count, speed and density, TTC and PET, and a crash prediction model with hotspot and level-of-service analysis. The governing equations are presented.

The work done so far is limited to a **sample UAV video**. A detection and tracking pipeline built on a pretrained YOLOv8 model and ByteTrack produced 19,133 tracking records for 614 vehicles in pixel coordinates, whereas a second pretrained model failed to identify motorcycles, showing that existing models are not reliable for local traffic. Trajectories in real-world coordinates have not been found, because ground coordinates are not yet available. The UAV (DJI Neo 2) has not been purchased and the mid-block sections are yet to be selected, so field data collection with DGPS control is pending.

Keywords: UAV, Vehicle Detection, YOLO, Small-Object Detection, Multi-Object Tracking, ByteTrack, Vehicle Trajectory, DGPS Georeferencing, Time-to-Collision, Post-Encroachment Time, Surrogate Safety Measures, Extreme Value Theory, Highway Traffic Safety

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CHAPTER-1
INTRODUCTION

1.1 GENERAL

Road traffic safety is a major concern at highway junctions and mid-block sections, where congestion, speed differences, lane changing and unsafe driving contribute to accident risk. Safety studies have traditionally depended on crash records, manual observations, roadside sensors and fixed cameras. Crash records are retrospective and sparse, while manual surveys and fixed sensors are costly and observe only the stretch of road within their range. For proactive safety analysis, the ability to reconstruct how every vehicle moves before a conflict occurs is particularly valuable.

Unmanned Aerial Vehicles (UAVs) offer a low-cost and flexible platform for capturing high-resolution traffic imagery over a wide road stretch. From the bird's-eye view of a UAV, several lanes and many interacting vehicles appear in the same frame, so that vehicle trajectories can be extracted rather than only point measurements such as flow or speed at a detector location. Trajectories of this kind can be converted into traffic parameters such as vehicle count, speed and density, and into surrogate safety measures such as time-to-collision (TTC) and post-encroachment time (PET). Because conflicts can be evaluated before a crash occurs, surrogate safety measures reduce the dependence on long crash histories (Arun et al., 2021).

Deep learning object detectors such as YOLO, combined with multi-object tracking, now make it possible to process long UAV videos automatically. Vehicles seen from a UAV, however, occupy only a few pixels, and two-wheelers are the smallest of them. This is a serious difficulty for Indian highways, where two-wheelers, three-wheelers, passenger cars, light commercial vehicles and heavy vehicles share the carriageway without strict lane discipline. A detector that is reliable on benchmark data may therefore not be reliable on local traffic.

This study proposes an improved YOLO-based detection and trajectory analysis framework for traffic safety monitoring, crash risk assessment and accident hotspot identification on mid-block highway sections in Thiruvananthapuram. Vehicles of five classes (two-wheeler, three-wheeler, car, light commercial vehicle and heavy commercial vehicle) are to be detected and tracked in UAV video, their trajectories georeferenced using DGPS ground control, and the

resulting trajectories used to derive traffic parameters, surrogate safety measures and a crash prediction model. This interim report presents the literature review, objectives, data requirements, methodology and the part results obtained so far from a sample UAV video.

1.2 NEED FOR THE STUDY

The increasing volume and diversity of traffic on urban highways has created a need for safety assessment that does not wait for crashes to accumulate. In the absence of detailed vehicle-level data:

- Manual surveys and fixed roadside sensors remain costly and limited in spatial coverage
- Interactions between neighbouring vehicles, such as lane changes and sudden braking, cannot be observed
- Detectors trained on international UAV benchmarks may fail on small two-wheelers and three-wheelers in local traffic
- Safety measures cannot be computed reliably because pixel movement has no physical scale without georeferencing

In Thiruvananthapuram there is no locally annotated, DGPS-georeferenced UAV vehicle dataset covering the five vehicle classes. The need for such a dataset, and for a detector tuned to it, was confirmed by the sample-video tests described in Chapter 6, in which pretrained models proved unreliable for two-wheelers.

1.3 PROBLEM STATEMENT

Existing UAV-based traffic studies rely mainly on generic detectors, optical-flow or clustering methods, and rarely pair a small-object-optimised detector with trajectory-level crash risk analysis. The challenge lies in detecting all five vehicle classes reliably from high-altitude UAV imagery, converting the detections into georeferenced trajectories, and using those trajectories to estimate surrogate safety measures and crash risk on mid-block highway sections under heterogeneous Indian traffic.

1.4 OBJECTIVES

- To acquire UAV imagery of highway segments and construct an annotated, DGPS-georeferenced vehicle dataset across five classes

- To implement a baseline YOLO detector
- To derive vehicle count, speed and traffic density from the detected and tracked vehicles
- To evaluate microscopic surrogate safety measures, time-to-collision and post-encroachment time, using UAV DGPS trajectory data
- To develop a crash prediction model and rank the severity of potential collisions based on the evaluated surrogate safety measures

1.5 SCOPE OF THE STUDY

The study is limited to mid-block road sections in Thiruvananthapuram city (the sections are yet to be selected) and focuses on:

- Five vehicle classes: two-wheeler, three-wheeler, car, LCV and HCV
- A single UAV (DJI Neo 2, planned) operated in nadir view with DGPS ground control points
- A baseline YOLO detector and an improved small-object YOLO detector, compared with Faster R-CNN and RT-DETR
- Multi-object tracking, georeferenced trajectories, and vehicle count, speed and density
- Time-to-collision, post-encroachment time, crash risk estimation, hotspot and level-of-service analysis

1.6 SIGNIFICANCE OF THE STUDY

This study contributes to:

- A locally relevant, DGPS-georeferenced UAV vehicle dataset for Thiruvananthapuram traffic
- Evidence on how far pretrained detectors can be trusted for two-wheeler-heavy mixed traffic
- Trajectory-level safety evidence for mid-block sections, a setting that remains largely unexplored
- A reproducible workflow from UAV video to crash risk and hotspot maps that can be used by highway safety managers

CHAPTER-2
LITERATURE REVIEW

2.1 REVIEW OF LITERATURE

1. Ke et al. (2017)

Ke et al. (2017) proposed a framework that combines a KLT tracker with k-means clustering to separate traffic streams from moving-camera UAV video. From the separated streams the method estimates speed, density and volume of each traffic stream in real time. The authors reported accuracies of 96% for average speed and 87% for vehicle count.

The study represents an early stage in the development of UAV traffic analysis: it showed that aerial video can supply traffic-flow parameters and not only visual records. It is relevant to the present work because the count, speed and density estimates planned in this study follow the same logic, and because comparison with reference counts and speeds is used for validation.

2. Khan et al. (2017)

Khan et al. (2017) reviewed the literature on UAV-based traffic studies and proposed a universal guiding framework of seven components: scope definition, flight planning, flight implementation, data acquisition, data processing and analysis, data interpretation, and optimised traffic application. The authors treated a UAV study as a complete data-acquisition system and not merely as video recording.

They emphasised that flight altitude, camera orientation, weather, lighting and battery limits all affect the quality of the final trajectories. For the present study this means that flight planning has to be designed around trajectory accuracy and not only around visual coverage.

3. Xu, Y. et al. (2017)

Xu et al. (2017) extended the Faster R-CNN framework to detect cars in low-altitude UAV imagery at signalised intersections. The study demonstrated robustness to changes in illumination and to in-plane rotation of vehicles, and observed that detection speed stayed nearly constant regardless of the number of vehicles in a frame.

Faster R-CNN is a two-stage detector in which a Region Proposal Network first generates candidate regions that a second stage classifies and localises. It can give strong localisation,

but its computational cost is a concern for long UAV videos. In the present study it serves as a two-stage benchmark against which the proposed YOLO detectors are compared.

4. Du et al. (2018)

Du et al. (2018) introduced UAVDT, a vehicle-focused aerial benchmark for object detection and tracking. The benchmark highlights high object density, small objects, camera motion and real-time constraints as the central challenges of UAV vision.

The study showed that generic object detectors can lose performance under UAV-specific conditions. UAVDT is valuable for pretraining and benchmarking, but it cannot substitute for a locally annotated dataset, which is the reason a site-specific five-class dataset is planned in this study.

5. Liu et al. (2020)

Liu et al. (2020) proposed UAV-YOLO, which modifies the YOLOv3 Darknet backbone to enlarge the receptive field and enrich spatial information for small objects seen from a UAV. The authors also built a dedicated UAV-viewed training dataset and a training method suited to it.

This is one of the foundational works on small-object detection tailored to the UAV viewpoint. It supports the view that the detector architecture should be adapted to the scale distribution of UAV imagery and informs the small-object head proposed in this study.

6. Outay et al. (2020)

Outay et al. (2020) reviewed the applications of UAVs in road safety, traffic monitoring and highway infrastructure management. The review discussed computer vision methods for traffic flow analysis, risk assessment and damage assessment, and identified the barriers to wide-scale UAV deployment, such as regulations, flight limitations and data processing demands, together with countermeasures.

The review provides the overall context for the present study and confirms the analytical opportunities of UAV data as well as the practical constraints, such as permissions, that must be cleared before field acquisition.

7. Arun et al. (2021)

Arun et al. (2021) carried out a systematic mapping review of 549 studies on surrogate safety assessment using traffic conflict techniques and developed a broad framework for it. They stressed the need for clearer relationships between conflicts and crashes and for a better understanding of how individual measures should be used.

The review shows that conflicts can be evaluated before a crash occurs, reducing dependence on long crash histories. It also cautions that surrogate measures identify conflict conditions and not confirmed crashes, so any crash model developed in this study is presented as estimated risk.

8. Aharon et al. (2022)

Aharon et al. (2022) proposed BoT-SORT, a tracker that combines motion and appearance information for data association and includes camera-motion compensation. Camera-motion compensation is important for UAV video, where even a small camera movement shifts all vehicles in the image.

A tracker that assumes a stationary camera may confuse camera motion with vehicle movement. BoT-SORT is therefore one of the trackers to be compared on identity continuity in this study.

9. Zhang et al. (2022)

Zhang et al. (2022) proposed ByteTrack, which associates every detection box, including low-confidence ones, instead of discarding them. This helps to recover partially occluded objects whose detector confidence temporarily drops, a situation common for small vehicles seen from above.

ByteTrack is the tracker used in the pipeline test on the sample video reported in Chapter 6, where it maintained vehicle identities across frames.

10. Zong et al. (2022)

Zong et al. (2022) combined deep-learning-based vehicle tracking with time-to-collision analysis on UAV video at road intersections. The study demonstrated UAV-derived trajectories as a practical tool for proactive collision risk assessment and highlighted the advantage of UAV observation in overcoming the occlusion that affects conventional tracking equipment.

The work is a direct example of the detection–tracking–safety chain that the present study aims to build, and it highlights the value of such analysis for urban road agencies and traffic safety management.

11. Wang et al. (2023)

Wang et al. (2023) proposed UAV-YOLOv8, a modification of YOLOv8 for small-object detection in UAV aerial photography scenarios. The work belongs to the recent line of YOLO research that focuses on preserving fine spatial information for tiny targets.

It supports adapting the detector architecture to the scale distribution of UAV imagery instead of using a generic model, which is the principle behind the improved detector of this study.

12. Berghaus et al. (2024)

Berghaus et al. (2024) built a drone-based trajectory dataset of over 8,000 trajectories on a German highway, covering an off-ramp, an on-ramp and congested traffic. The pipeline used YOLOv5 detection with three-dimensional camera calibration to project vehicles onto the road surface, followed by post-processing of the trajectories.

The derived speed and acceleration were compared with independent sources, and the data were applied to traffic-flow and accident-risk analysis. The study shows that calibration and quality assessment must be planned as part of the measurement chain, and that a trajectory dataset is a research output in its own right.

13. Xu, L. et al. (2024)

Xu et al. (2024) improved YOLOv8n with multi-scale feature fusion, conditionally parameterised convolutions and the Wise-IoU loss for small-object detection in UAV images. The model achieved state-of-the-art accuracy on the VisDrone benchmark for small, partially occluded objects.

It outperformed other lightweight YOLO variants and Faster R-CNN on small-object accuracy while remaining lightweight. The multi-scale fusion and Wise-IoU components are among the changes to be tested in the improved detector of this study.

14. Zhao et al. (2024)

Zhao et al. (2024) proposed RT-DETR, a real-time end-to-end transformer detector that removes the non-maximum suppression post-processing step. The RT-DETR-R50 model

reached 53.1% AP on COCO at 108 frames per second on a T4 GPU, outperforming advanced YOLO detectors in both speed and accuracy.

RT-DETR serves, together with Faster R-CNN, as a comparison detector against which the proposed detector is benchmarked.

15. Fonod et al. (2025)

Fonod et al. (2025) developed a multi-drone pipeline for extracting georeferenced vehicle trajectories, demonstrated over 20 intersections. Vehicle detections were used during image registration so that moving vehicles do not dominate the estimation of camera motion, and trajectories were stabilised before being georeferenced through orthophoto and master-frame alignment.

The study demonstrates that trajectory extraction and georeferencing should be considered together, and it supports the DGPS georeferencing step and the frame-stabilisation approach planned in this study.

16. Kumar and Sarkar (2025)

Kumar and Sarkar (2025) collected UAV-based vehicle trajectory data at the Panipat Toll Plaza on NH-44, an Indian highway toll plaza with heterogeneous, lane-changing traffic. Lane-Changing Time-to-Collision, a surrogate safety measure, was combined with Extreme Value Theory (EVT) to estimate crash risk.

The authors found a negative relationship between vehicle size and conflict involvement across vehicle categories. The study is closely aligned with the present work because it connects UAV data, heterogeneous Indian traffic, lane changing and crash-risk estimation, and it shows why class-wise analysis is essential.

17. Maji and Ghosh (2025)

Maji and Ghosh (2025) extracted trajectories from high-resolution drone video at roundabouts under non-lane-based mixed traffic. TTC and maximum Delta-V were used as surrogate safety measures, and Extreme Value Theory was applied for conflict analysis, moving beyond simple threshold-based conflict counting.

The study is important for Indian applications because road-user interactions do not always follow strict lane-based assumptions, and it supports the use of statistical modelling of extreme conflicts in this study.

2.2 SUMMARY OF LITERATURE

The reviewed studies show that UAVs are well established as a low-cost, flexible platform for road safety, traffic monitoring and highway infrastructure surveys (Outay et al., 2020; Khan et al., 2017). Classical tracking and clustering methods (KLT with k-means) and two-stage detectors (Faster R-CNN) have both demonstrated accurate traffic parameter and vehicle-count estimation from aerial video (Ke et al., 2017; Xu et al., 2017).

Standard YOLO detectors are fast, but recent studies (UAV-YOLO, UAV-YOLOv8, YOLOv8n with multi-scale fusion) show that unmodified YOLO underperforms on small, high-altitude vehicle objects, and benchmarks such as UAVDT confirm that generic detectors lose accuracy under UAV conditions (Liu et al., 2020; Wang et al., 2023; Xu et al., 2024; Du et al., 2018). Tracking methods such as ByteTrack and BoT-SORT address identity continuity, and BoT-SORT adds camera-motion compensation that is relevant to UAV video (Zhang et al., 2022; Aharon et al., 2022).

UAV-derived vehicle trajectories have been used successfully with surrogate safety measures such as time-to-collision to assess crash and conflict risk at intersections, toll plazas and roundabouts, including on Indian roads (Zong et al., 2022; Kumar and Sarkar, 2025; Maji and Ghosh, 2025). Trajectory datasets and georeferencing pipelines show that calibration, post-processing and independent validation are essential (Berghaus et al., 2024; Fonod et al., 2025). Very few studies, however, combine an improved YOLO-based detector with trajectory-level crash risk and hotspot analysis for Indian mixed-traffic conditions.

2.3 IDENTIFIED RESEARCH GAPS

Based on the review of literature, the following research gaps have been identified:

2.3.1 Generic detectors instead of small-object-optimised YOLO

Existing UAV traffic studies rely mainly on generic detectors, optical-flow or clustering methods rather than modern small-object-optimised YOLO architectures.

2.3.2 Computational cost of two-stage detectors

Two-stage detectors such as Faster R-CNN achieve good small-object accuracy but are computationally heavy for near real-time, highway-scale processing.

2.3.3 Lack of evaluation on Indian mixed vehicle classes

Recent small-object YOLO improvements have not been evaluated on Indian mixed vehicle-class highway traffic (two-wheeler, three-wheeler, car, LCV, HCV).

2.3.4 Detection and safety analysis treated separately

UAV-trajectory-based crash risk and surrogate safety studies exist, but are rarely paired with an improved, small-object-optimised detector or extended into hotspot and LOS-based safety applications.

2.3.5 Mid-block sections largely unexplored

UAV-based, trajectory-level traffic safety analysis for mid-block sections remains largely unexplored.

2.4 RESEARCH CONTRIBUTION

The present study addresses the identified gaps through the following contributions:

- Acquisition of UAV imagery and construction of an annotated, DGPS-georeferenced vehicle dataset with five vehicle classes
- A baseline YOLO detector and an improved small-object-optimised YOLO detector, benchmarked against Faster R-CNN and RT-DETR on the same local test set
- Multi-object tracking and georeferenced trajectories from which vehicle count, speed and density are derived and validated
- Surrogate safety measures (TTC and PET) and a crash prediction model with severity ranking of potential collisions
- Accident hotspot identification and level-of-service assessment presented through GIS visualisation

2.5 CONCLUSION OF LITERATURE REVIEW

The literature establishes that UAV imagery, deep-learning detection, multi-object tracking and surrogate safety measures can be combined into a complete workflow from video to crash

risk. It also shows that small and partially occluded vehicles, camera motion, mixed traffic, trajectory noise and georeferencing remain open challenges, and that local annotated data are needed because models do not transfer reliably between locations.

The present study builds on these methods while addressing the gaps through a locally annotated, DGPS-referenced five-class dataset, a small-object-optimised detector and trajectory-level safety analysis. Its evaluation covers not only detector accuracy but also the quality and safety relevance of the resulting trajectories.

CHAPTER-3
STUDY AREA

3.1 GENERAL

The study is set in Thiruvananthapuram city, Kerala, India. Thiruvananthapuram is the capital of the state of Kerala and a rapidly urbanising centre with a mix of residential, commercial, institutional and administrative land uses. This land-use mix, together with major road corridors, produces a traffic stream in which two-wheelers, three-wheelers, cars, light commercial vehicles and heavy commercial vehicles all share the road. This mixed vehicle composition makes the city a suitable setting for testing UAV-based detection and trajectory analysis under heterogeneous traffic.

3.2 LOCATION AND EXTENT

Thiruvananthapuram Corporation is located in the southern part of Kerala and serves as the administrative and economic centre of the region. The corporation covers an area of approximately 214.86 square kilometres, is divided into 100 wards, and is bounded by the Arabian Sea on the west. Figure 3.1 shows the location of Thiruvananthapuram Corporation within Kerala and India.

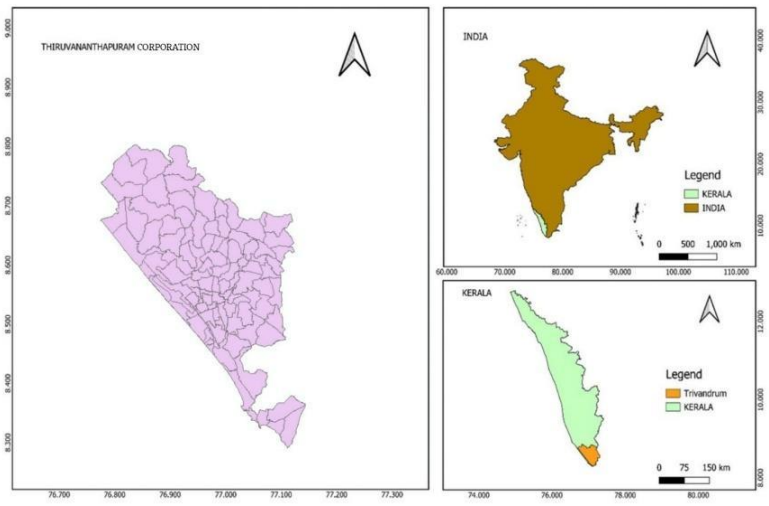


Figure 3.1: Study Area

3.3 SELECTION OF MID-BLOCK STUDY SECTIONS

The survey will be carried out on mid-block road sections of the city. **No mid-block road section has been shortlisted or confirmed so far.** Selecting the sections is the first pending step of the project, because flight planning, DGPS ground control and data collection all depend on it. The sections will be chosen using the criteria in Table 3.1.

Table 3.1: Proposed criteria for selecting mid-block sections

Criterion	Requirement
Mixed traffic	2W, 3W, cars, LCV and HCV all present
Steady flow	Adequate, consistent volume during the survey
Simple geometry	Straight, level mid-block with clear lane markings
Safety relevance	Lane changes, speed differences and sudden braking
UAV feasibility	Open sky, safe take-off point and room for ground control points
Permissions	Aviation, institutional and privacy clearance

UAV data acquisition with DGPS ground control will follow once the sections are selected and the UAV (DJI Neo 2) is purchased.

3.4 RELEVANCE OF THE STUDY AREA

Thiruvananthapuram is considered suitable for the study for the following reasons:

- Mixed vehicle composition that includes all five vehicle classes of interest
- Diverse land use, which generates varied traffic demand along the road network
- Mid-block sections where lane changing, speed differences and sudden braking are the main conflict mechanisms
- A need for a local, georeferenced dataset because published Indian studies of this kind are few

3.5 SUMMARY

The study area is Thiruvananthapuram city. The mid-block sections are still to be selected using the criteria in Table 3.1, and all field activities depend on this selection.

CHAPTER-4
DATA AND SOFTWARE USED

4.1 GENERAL

The accuracy of UAV-based traffic analysis depends on the quality of the imagery, the ground control and the annotations. This chapter lists all the data required to achieve the objectives, together with the type, source, technical details and status of each dataset, followed by the UAV platform and the software used.

4.2 DATA REQUIRED

Table 4.1 summarises the data. Datasets marked "To be collected" cannot be obtained until the UAV is purchased and the mid-block sections are selected.

Table 4.1: Data requirement and status

Data	Type / technical details	Source	Status
UAV video of mid-block sections	MP4 video, nadir view; DJI Neo 2 records up to 4K/100 fps	Field survey using DJI Neo 2	To be collected (UAV not yet purchased; sections not yet selected)
DGPS ground control points	Point coordinates (X, Y, Z) of GCPs marked on the road surface	Field DGPS survey	To be collected
Annotated vehicle dataset	Bounding boxes, 5 classes (2W, 3W, car, LCV, HCV), YOLO format	CVAT annotation of UAV frames	To be prepared after acquisition
Road and lane geometry	Vector shapefiles / raster orthomosaic	Digitised from UAV orthomosaic	To be prepared after acquisition
Reference counts, speeds, crash / conflict records	Tabular data for validation	Manual counts from video; local records	To be collected
Sample UAV video	MP4, about 1280 × 736 px, 24–25 FPS, 477 frames	Existing sample video	Available; used for pipeline testing
Sample tracking output	Pixel-coordinate table (frame, time, track ID, class, confidence, box); 19,133 records, 614 track IDs	YOLO + ByteTrack on the sample video	Generated (pixel domain; not georeferenced)

4.2.1 UAV Video

Nadir (top-down) video of the selected mid-block sections will be recorded with the DJI Neo 2 at a fixed altitude. The video is the primary input for detection, tracking and trajectory extraction. Overlapping stills from the same flights are used to build the orthomosaic.

4.2.2 DGPS Ground Control Points

Ground control points (GCPs) will be marked on the road surface and surveyed with DGPS to obtain their (X, Y, Z) coordinates. At least four are required to georeference the trajectories; additional surveyed points are retained as independent check points.

4.2.3 Annotated Vehicle Dataset

Frames extracted from the UAV video will be labelled in CVAT with bounding boxes for the five vehicle classes and exported in YOLO format. This dataset is used to train and test the baseline and improved detectors.

4.2.4 Road and Lane Geometry

Lane lines, road edges and centrelines will be digitised from the orthomosaic as vector layers. They are used for lane assignment, for checking projected trajectories against the lane boundaries and for mapping conflicts.

4.2.5 Reference and Validation Data

Manual vehicle counts and speeds taken from the same video, and local crash or conflict records where available, are used to validate the estimated count, speed, density and safety measures.

4.2.6 Sample UAV Video

A sample UAV video, available before field acquisition, was used to test the detection and tracking pipeline. Its use and results are described in Chapter 6.

4.3 UAV PLATFORM

The planned platform is the DJI Neo 2 (Figure 4.1). **The drone has not yet been purchased**, so UAV data acquisition with DGPS ground control has not yet taken place. Its specifications are given in Table 4.2. With a flight time of about 19 minutes per battery, several sorties will

be needed across peak and off-peak periods, and altitude, time and weather will be logged for each sortie.



Figure 4.1: DJI Neo 2 (planned UAV platform)

Table 4.2: Specifications of DJI Neo 2

Parameter	Specification
Weight	151 g (160 g with O4 transceiver)
Camera	12 MP, 1/2-inch CMOS sensor
Video	Up to 4K/100 fps, 2-axis mechanical gimbal with EIS
Flight time	Approximately 19 minutes
Max transmission range	6 km (CE) / 10 km (FCC)
Obstacle sensing	Omnidirectional: vision, LiDAR and infrared

4.4 DATA PRE-PROCESSING

The following pre-processing steps are planned once the field data are available:

- Extraction of every k-th frame for annotation to avoid near-duplicate images
- Stabilisation of the video to a reference frame, and removal of blurred or unusable frames
- Generation of the orthomosaic and digitisation of road geometry
- Export of annotations in YOLO format, with train, validation and test sets split by sortie or location
- Conversion of georeferenced coordinates to a metric projected system (for example UTM zone 43N, EPSG:32643)

4.5 SOFTWARE USED

The following software and computing environment were used in the sample-video test, or are planned for later stages:

- **Google Colab** with an NVIDIA Tesla T4 GPU for model inference, with the video stored on Google Drive
- **Python, PyTorch and OpenCV** for video handling and processing
- **Ultralytics YOLO** (YOLOv8) for detection, and **ByteTrack** for multi-object tracking
- **pandas** for building the trajectory table from the tracker output
- **CVAT** for annotation of the five vehicle classes (planned)
- Photogrammetry software for the orthomosaic and GIS software for mapping, hotspot and LOS analysis (to be finalised)

4.6 DATA LIMITATIONS

The study currently faces the following limitations related to data:

- The UAV has not been purchased, so no field imagery or DGPS coordinates exist yet
- The mid-block sections are not yet selected, so site-specific flight planning cannot be finalised
- Only a sample video is available; its results cannot be generalised to the study sections
- Flight time of about 19 minutes per battery limits the length of each observation
- Cars are expected to outnumber heavy vehicles in the annotated data, which will lead to class imbalance

These limitations are to be addressed by selecting the sections first, purchasing the UAV, planning sorties across peak and off-peak periods, and using class weights or oversampling during training.

CHAPTER-5
METHODOLOGY

5.1 GENERAL

The methodology is a UAV-based workflow that moves from field acquisition to detection, tracking, georeferenced trajectories and finally to traffic parameters and crash risk. Each step uses data from the previous one, so errors in an early stage can propagate: a missed vehicle breaks a trajectory, an identity switch joins two vehicles, and a coordinate error changes the relative position of two vehicles and therefore their TTC or PET. Detection, tracking, georeferencing and post-processing are therefore treated as one connected system. The sample-video work reported in Chapter 6 does not change this methodology.

5.2 OVERALL METHODOLOGICAL FRAMEWORK

The study follows the sequence of nine stages shown in Figure 5.1, preceded by the selection of the mid-block sections and followed by crash prediction, hotspot and LOS analysis and benchmarking against Faster R-CNN, RT-DETR and manual counts.

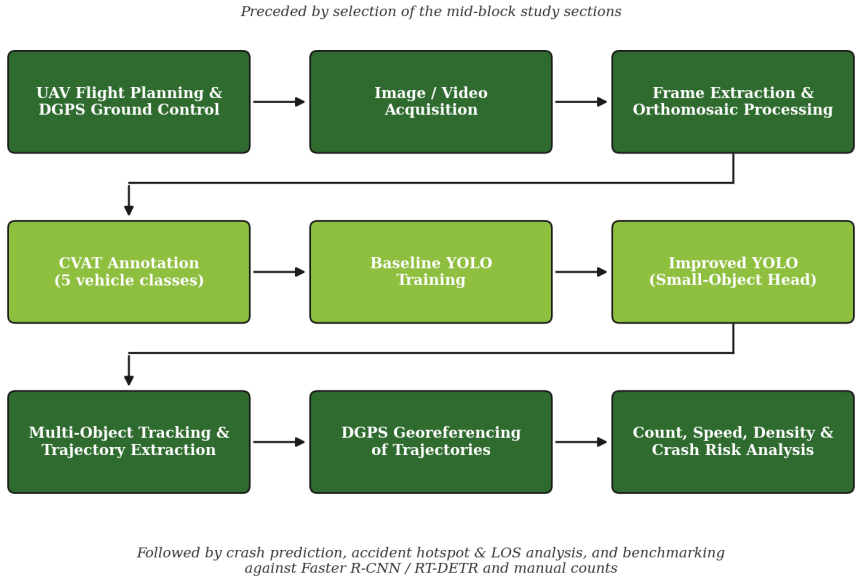


Figure 5.1: Methodology flowchart

The twelve methodological steps described in this chapter are:

1. Selection of the mid-block study sections (Section 5.3)
2. UAV flight planning and DGPS ground control (Section 5.4)
3. Video and image acquisition (Section 5.5)
4. Frame extraction, stabilisation and orthomosaic processing (Section 5.6)
5. CVAT annotation of five vehicle classes (Section 5.7)
6. Baseline YOLO training (Section 5.8)
7. Improved small-object YOLO (Section 5.9)
8. Multi-object tracking and trajectory extraction (Section 5.10)
9. DGPS georeferencing of trajectories (Section 5.11)
10. Count, speed and density (Section 5.12)
11. Surrogate safety measures: TTC and PET (Section 5.13)
12. Crash risk estimation and prediction, hotspot, LOS and benchmarking (Sections 5.14 and 5.15)

5.3 SELECTION OF MID-BLOCK STUDY SECTIONS

The flight plan can be fixed only after a section is chosen, because it depends on road width, traffic density and the expected vehicle classes. Sections are shortlisted using the criteria of Table 3.1 (mixed traffic, steady flow, simple geometry, safety relevance, UAV feasibility and permissions). Aviation, institutional and privacy permissions are verified before any flight.

5.4 UAV FLIGHT PLANNING AND DGPS GROUND CONTROL

The flight altitude is set from the ground sampling distance (GSD), so that the smallest class, the two-wheeler, still spans enough pixels. A higher flight covers more road but makes vehicles smaller, while a lower flight improves object size at the cost of coverage and more perspective distortion (Khan et al., 2017). The GSD is

$$\text{GSD} = \frac{p \times H}{f}$$

Where:

- GSD = ground distance covered by one image pixel
- p = physical size of one sensor pixel
- H = flight height above the road surface
- f = focal length of the camera

A nadir (top-down) view is used, with the UAV hovering over the road centreline, to limit perspective distortion; the view must keep every vehicle in frame long enough to form a trajectory. Ground control points are marked on the road surface and surveyed with DGPS in (X, Y, Z). At least four are needed; extra points are kept as independent check points. Berghaus et al. (2024) and Fonod et al. (2025) show that calibration and georeferencing must be planned as part of the measurement chain.

5.5 VIDEO AND IMAGE ACQUISITION

Nadir 4K video is recorded with the DJI Neo 2 at a fixed altitude, in good light and low wind. A position is recorded once per frame, so the movement between two frames equals speed divided by frame rate:

$$\Delta t = \frac{1}{f_r}, \qquad d = v \times \Delta t$$

Where:

- Δt = time between two frames (s)
- f_r = video frame rate (frames per second)
- d = distance moved by a vehicle between two frames (m)
- v = vehicle speed (m/s)

A low frame rate gives large jumps that amplify noise in speed and acceleration, whereas a high frame rate raises storage and inference cost. IoU-based tracking also needs consecutive boxes to overlap. For two equal boxes of length l along the direction of motion that are shifted by d, the overlap is

$$\text{IoU} = \frac{l - d}{l + d}$$

For example, at 60 km/h a vehicle moves 0.67 m per frame at 25 fps and 0.33 m at 50 fps. With assumed sizes, a 4.5 m car keeps about 74% overlap at 25 fps, whereas a 2 m two-wheeler keeps only 50%, rising to 71% at 50 fps. Short vehicles such as two-wheelers therefore lose overlap fastest, which favours a higher frame rate. Because the flight time is about 19 minutes per battery, several sorties are planned across peak and off-peak periods, and altitude, time and weather are logged for each. The output is raw MP4 video per sortie plus overlapping stills for

the orthomosaic. Ke et al. (2017) showed that speed, density and volume can be estimated from aerial video.

5.6 FRAME EXTRACTION, STABILISATION AND ORTHOMOSAIC

Every k-th frame is extracted for annotation to avoid near-duplicate images, whereas detection and tracking run on the complete video, as in the sample-video test. To stop camera motion from being mistaken for vehicle motion, features are matched between consecutive frames, a homography is estimated with RANSAC and every frame is warped to a reference frame. Detected vehicles are masked out of the feature matching so that moving traffic does not bias the camera-motion estimate, the approach followed by Fonod et al. (2025).

Overlapping stills are processed by photogrammetry (structure from motion) into an orthomosaic, from which lane lines, road edges and centrelines are digitised as road geometry. Blurred or unusable frames are removed before annotation.

5.7 CVAT ANNOTATION OF FIVE VEHICLE CLASSES

Five vehicle classes are used (Table 5.1). Each vehicle receives a tight bounding box, with the rider included for two-wheelers, and written rules for partly visible and occluded vehicles are fixed before labelling starts. Annotations are exported in YOLO format, one line per box with normalised coordinates. Vehicle category matters for conflict analysis in Indian traffic (Kumar and Sarkar, 2025), so class-wise labels are essential.

Table 5.1: Vehicle classes used for annotation

Code	Class	Typical examples
2W	Two-wheeler	Motorcycle and scooter, rider included
3W	Three-wheeler	Auto-rickshaw
Car	Car	Passenger car
LCV	Light commercial vehicle	Small goods vehicle and van
HCV	Heavy commercial vehicle	Truck and bus

The train, validation and test sets are split by sortie or location, not by random frame, because neighbouring frames are near-duplicates and would leak into the test set. Cars will outnumber heavy vehicles, so class weights or oversampling are used and results are reported class by class. A second person re-checks a sample of the labels; agreement of boxes is measured by IoU and agreement of classes by Cohen's kappa:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

Where:

- p_o = observed agreement between the two annotators
- p_e = agreement expected by chance

5.8 BASELINE YOLO TRAINING

A YOLOv8 model (Ultralytics), the family used in the sample-video test, starts from pretrained weights and is fine-tuned on the local five-class dataset. The baseline fixes the reference accuracy against which the small-object model is judged on the same held-out test set. Training uses an input size of around 960 px (as in the sample test, to be tuned), data augmentation and early stopping on validation mAP, and all hyper-parameters are logged. The total training loss is the weighted sum of a box regression loss, a classification loss and a distribution focal loss:

$$L = \lambda_{box} L_{CIoU} + \lambda_{cls} L_{BCE} + \lambda_{dfl} L_{DFL}$$

The box loss uses the complete IoU (CIoU), which adds a centre-distance term and an aspect-ratio term to the overlap term:

$$L_{CIoU} = 1 - \mathrm{IoU} + \frac{\rho^2(b, b_{gt})}{c^2} + \alpha v$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w_{gt}}{h_{gt}} - \arctan\frac{w}{h}\right)^2, \qquad \alpha = \frac{v}{(1 - \mathrm{IoU}) + v}$$

Where:

- IoU = Area(A ∩ B) / Area(A ∪ B), the overlap between predicted box A and ground-truth box B
- ρ = distance between the centres of the predicted box b and ground-truth box b_gt
- c = diagonal length of the smallest box enclosing both boxes
- w, h = width and height of the predicted box (subscript gt for the ground truth)

Detection performance is reported using precision, recall, F1 score and mean average precision (mAP), together with the confusion matrix, class-wise AP and inference speed in frames per second:

$$\text{Precision} = \frac{TP}{TP+FP}, \qquad \text{Recall} = \frac{TP}{TP+FN}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$AP = \sum_{k} (r_k - r_{k-1})\, p_{interp}(r_k), \qquad \text{mAP} = \frac{1}{N} \sum_{i=1}^{N} AP_i$$

Where:

- TP, FP, FN = true positives, false positives and false negatives (a detection is a true positive when its IoU with a ground-truth box exceeds the chosen threshold)
- p_{interp} = interpolated precision at recall r
- N = number of classes

5.9 IMPROVED SMALL-OBJECT YOLO

A feature-map cell at stride s covers s × s pixels of the image, so a small vehicle fills only a small fraction of one cell and its features are lost. For an object of width w and height h pixels, the fraction of one cell it covers is

$$\text{Fraction of cell covered} = \frac{w \times h}{s^2}$$

For example, a two-wheeler 16 px wide and 16 px tall at stride 32 fills only 256/1024 = one quarter of a cell. The following changes are tested to improve small-vehicle detection:

- A high-resolution stride-4 (P2) detection head
- Multi-scale feature fusion
- The Wise-IoU box loss, which weights the IoU loss according to the quality of each box (Xu et al., 2024)
- Scale and mosaic augmentation

An ablation is carried out in which one change is added at a time to show which one raises small-vehicle AP. The baseline and improved models use the identical held-out local test set, and both accuracy and speed are reported. The improved model is also compared with Faster

R-CNN and RT-DETR. Liu et al. (2020) enlarged the receptive field for small UAV objects, and Xu et al. (2024) used multi-scale fusion and Wise-IoU on YOLOv8n.

5.10 MULTI-OBJECT TRACKING AND TRAJECTORY EXTRACTION

The tracker takes the per-frame boxes, class labels and confidence scores from the detector. A Kalman filter for each vehicle predicts its box in the next frame, assuming constant velocity, and is corrected by the matched detection. With state vector x (box centre, box size and their velocities) and measurement z, the prediction and update steps are:

$$x_{k|k-1} = F\,x_{k-1|k-1}, \qquad P_{k|k-1} = F\,P_{k-1|k-1}F^T + Q$$

$$K_k = P_{k|k-1}H^T\left(H\,P_{k|k-1}H^T + R\right)^{-1}$$

$$x_{k|k} = x_{k|k-1} + K_k\left(z_k - H\,x_{k|k-1}\right)$$

$$P_{k|k} = (I - K_kH)\,P_{k|k-1}$$

Where:

- F = state transition matrix (constant-velocity model)
- H = measurement matrix
- Q, R = process and measurement noise covariances
- K_k = Kalman gain at frame k

In ByteTrack (Zhang et al., 2022), high-confidence detections are matched to the predicted tracks first, using an IoU-based cost and the Hungarian algorithm. Leftover tracks are then matched to low-confidence detections, which recovers partly occluded vehicles. The cost of matching track i to detection j, and the assignment, are:

$$C_{ij} = 1 - \mathrm{IoU}(B_i, D_j), \qquad \min\sum_{i,j} C_{ij}\,a_{ij}$$

Where:

- B_i = predicted box of track i (from the Kalman filter)
- D_j = detected box j
- a_{ij} = 1 if track i is assigned to detection j, otherwise 0 (each track and detection used at most once)

ByteTrack, DeepSORT and BoT-SORT are compared on identity continuity; BoT-SORT adds camera-motion compensation (Aharon et al., 2022). Each persistent ID gives a time-ordered

sequence of positions. Short, fragmented or physically impossible tracks are repaired or rejected, because raw tracker output should not automatically be treated as the final trajectory. The anchor point of each box is its bottom centre, the point where the vehicle touches the road:

$$c_x = \frac{x_1 + x_2}{2}, \qquad c_y = y_2$$

Where:

- x_1, y_1, x_2, y_2 = top-left and bottom-right corners of the bounding box (pixels)

5.11 DGPS GEOREFERENCING OF TRAJECTORIES

Ground coordinates are needed before speed, distance, TTC or PET have any meaning, because pixel movement depends on camera geometry. This step cannot be carried out until DGPS coordinates exist. The road is treated as a plane, so an image point (u, v) maps to a ground point (E, N) by a homography estimated from at least four GCPs:

$$E = \frac{h_{11}u + h_{12}v + h_{13}}{h_{31}u + h_{32}v + h_{33}}$$

$$N = \frac{h_{21}u + h_{22}v + h_{23}}{h_{31}u + h_{32}v + h_{33}}$$

The matrix of coefficients h has eight independent values once it is normalised, and each GCP provides two equations, so four GCPs give a solution that is found by the direct linear transform and least squares when more points are used. Coordinates are expressed in a metric projected system (for example UTM zone 43N) and converted to latitude and longitude for GIS mapping. The accuracy of the transformation is judged on independent check points and fixed road features by the root mean square error:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left[(E_i - E_i')^2 + (N_i - N_i')^2\right]}$$

Where:

- E_i, N_i = surveyed coordinates of check point i
- E'_i, N'_i = coordinates of the same point obtained from the homography
- n = number of check points

Trajectories are also overlaid on the lane boundaries of the orthomosaic. A documented smoothing filter (for example Savitzky–Golay or a Kalman smoother) is applied with a window

short enough to keep real braking events, because direct differentiation amplifies measurement noise. Speed and acceleration are obtained from central differences of the smoothed positions:

$$v_i = \frac{p_{i+1} - p_{i-1}}{2\Delta t}, \qquad a_i = \frac{p_{i+1} - 2p_i + p_{i-1}}{\Delta t^2}$$

$$\text{Speed} = \sqrt{v_E^2 + v_N^2}$$

The output is a table of ID, time, image position (u, v), ground position (E, N), class, speed, acceleration and lane. Berghaus et al. (2024) used three-dimensional camera calibration and checked speed against other sources, and Fonod et al. (2025) stabilised trajectories before georeferencing.

5.12 COUNT, SPEED AND DENSITY

A virtual line across each lane counts every track ID once. Counts by class give the vehicle composition and the flow q. The speed of each vehicle comes from the smoothed, georeferenced trajectories. Space-mean speed (total distance over total time), and not the plain average of spot speeds, is used in flow relations. For a section of length L observed for time T, with N vehicles counted:

$$q = \frac{N}{T}, \qquad k = \frac{N}{L}$$

$$v_s = \frac{\sum_i d_i}{\sum_i t_i}, \qquad q = k \times v_s$$

Where:

- q = flow (vehicles per hour)
- k = density (vehicles per km), measured directly from the UAV view
- v_s = space-mean speed
- d_i, t_i = distance travelled and time spent by vehicle i within the section

Edie's generalised definitions use the whole space–time region L × T, with flow q equal to the sum of the distances travelled divided by L × T and density k equal to the sum of the times spent divided by L × T. Lane assignment and lane changes follow from the digitised lane geometry. PCU factors can be applied to the counts for the level-of-service analysis. Estimates are compared with manual counts and speeds from the same video using the percentage error:

$$\text{Error}\,(\%) = \frac{|N_{est} - N_{ref}|}{N_{ref}} \times 100$$

5.13 SURROGATE SAFETY MEASURES: TTC AND PET

Safety measures are computed for leader–follower pairs in the same lane or path. Lane-change and merging interactions are handled separately. TTC is the time to collision if both vehicles keep their current speeds; it is defined only when the follower is faster than the leader. It is computed every frame, and the minimum of each interaction is kept as its severity:

$$\text{TTC} = \frac{x_L - x_F - l_L}{v_F - v_L}, \qquad v_F > v_L$$

$$\text{TTC}_{min} = \min_t \text{TTC}(t)$$

Where:

- x_L, x_F = positions of the leader and the follower along the direction of travel
- v_L, v_F = speeds of the leader and the follower
- l_L = length of the leader

For crossing or merging paths, post-encroachment time (PET) is the gap between the first vehicle leaving the conflict point and the second vehicle arriving at it. As an optional complement, the deceleration rate to avoid a crash (DRAC) gives the deceleration the follower needs:

$$\text{PET} = t_2 - t_1$$

$$\text{DRAC} = \frac{(v_F - v_L)^2}{2\,(x_L - x_F - l_L)}$$

Where:

- t_1 = time at which the first road user leaves the conflict point
- t_2 = time at which the second road user arrives at the conflict point

No single threshold is universal, so the thresholds of TTC and PET are tested over a range and the sensitivity of the conflict counts is reported. Conflicts are ranked by minimum TTC and PET, by vehicle-class pair and by location. TTC should be interpreted within a defined prediction horizon and not as a universal crash predictor. The measures follow Arun et al. (2021), and UAV trajectories were used with TTC-type measures by Zong et al. (2022) and Kumar and Sarkar (2025).

5.14 CRASH RISK ESTIMATION AND PREDICTION

Crashes are rare, so risk is estimated from the extreme tail of the conflict distribution using Extreme Value Theory. Minimum TTC values are negated so that the most severe conflicts become the maxima, Y = −TTCmin. In the block-maxima approach, the most severe value in each time block is fitted with a Generalised Extreme Value (GEV) distribution:

$$G(y) = \exp\left\{-\left[1+\xi\left(\frac{y-\mu}{\sigma}\right)\right]^{-1/\xi}\right\}, \qquad 1+\xi\frac{y-\mu}{\sigma} > 0$$

Where:

- μ = location parameter
- σ = scale parameter ($\sigma > 0$)
- ξ = shape parameter ($\xi = 0$ gives the Gumbel distribution)

As an alternative, a Generalised Pareto distribution is fitted to the values beyond a threshold u (peaks over threshold):

$$H(y) = 1 - \left[1+\xi\frac{y-u}{\sigma}\right]^{-1/\xi}$$

The risk is the probability that the extreme passes zero ($TTC \leq 0$, which is a collision), reported by vehicle class and location:

$$\text{Risk} = P(Y \geq 0) = 1 - G(0)$$

Kumar and Sarkar (2025) combined lane-changing TTC with EVT on UAV data from an Indian toll plaza, and Maji and Ghosh (2025) used EVT for mixed-traffic roundabouts. A binary logistic regression model is also developed to predict conflict-prone events from the speed difference, density, minimum gap and class pair, and is used only if enough conflicts are observed:

$$P(\text{conflict} = 1) = \frac{1}{1+\exp\left[-\left(\beta_0 + \sum_{j=1}^{m}\beta_j x_j\right)\right]}$$

Where:

- β_0, β_j = intercept and regression coefficients
- x_j = explanatory variables (speed difference, density, minimum gap, vehicle-class pair)

The model is checked with a confusion matrix and the area under the ROC curve (ROC-AUC). Its output is estimated risk from surrogate indicators and is not a prediction of individual crashes. Conflicts are also ranked into severity classes by minimum TTC and PET to rank the severity of potential collisions.

5.15 HOTSPOT, LOS AND BENCHMARKING

Each conflict keeps its (E, N) location and is stored in GIS with lane geometry, section and vehicle-class attributes. Kernel density estimation shows where conflicts concentrate:

$$f(s) = \frac{1}{nh^2} \sum_{i=1}^{n} K\left(\frac{s - s_i}{h}\right)$$

Where:

- $f(s)$ = estimated conflict density at location s
- K = kernel function
- h = bandwidth
- s_i = location of conflict i

The Getis–Ord Gi* statistic tests whether the clusters are statistically significant:

$$G_i^* = \frac{\sum_j w_{ij} x_j - \bar{X} \sum_j w_{ij}}{S \sqrt{\frac{n \sum_j w_{ij}^2 - \left(\sum_j w_{ij}\right)^2}{n-1}}}$$

Where:

- x_j = conflict count in spatial unit j
- w_{ij} = spatial weight between units i and j
- $\bar{X}$, S = mean and standard deviation of x over all n units

Level of service (LOS) is obtained from the volume-to-capacity ratio in PCU and from density, using the applicable guideline classes (to be confirmed):

$$Q_{pcu} = \sum_c n_c e_c, \qquad v/c = \frac{Q_{pcu}}{C}$$

Where:

- n_c = number of vehicles of class c
- e_c = PCU factor of class c
- C = capacity of the section (PCU per hour)

Finally, the improved YOLO is benchmarked against the baseline YOLO, Faster R-CNN (Xu et al., 2017) and RT-DETR (Zhao et al., 2024) on the same test set using mAP and FPS, and counts and speeds are compared with manual reference values. An end-to-end check is made that better detection gives better tracks, trajectories, safety measures and hotspots, with each link evaluated (Table 5.2).

Table 5.2: Evaluation metrics adopted at each stage

Stage	Metrics
Detection	Precision, recall, F1, mAP, class-wise AP and confusion matrix, inference speed (FPS)
Tracking	Identity switches, track fragmentation and continuity; MOTA and IDF1
Trajectory	Positional error (RMSE on check points), speed error, plausibility of acceleration, continuity
Traffic parameters	Count error, speed error and density error against manual reference values
Safety	Agreement of TTC and PET conflicts with manually interpreted conflicts, stability of risk classification, spatial consistency

CHAPTER-6
RESULTS AND DISCUSSION

6.1 GENERAL

This chapter presents the part results obtained so far. Field work has not started: the UAV has not been purchased and the mid-block sections have not been selected. All results therefore come from a sample UAV video that was used to test the detection and tracking pipeline before field deployment. The results are discussed in order, from the work completed, through detection and tracking, to the reliability check of a second pretrained model and the current status of trajectory generation.

6.2 WORK COMPLETED TILL DATE

- The literature on UAV-based detection, tracking, georeferencing and surrogate safety measures has been reviewed (17 studies, Chapter 2)
- The UAV platform has been identified (DJI Neo 2) but not purchased, so UAV data acquisition with DGPS ground control has not yet taken place
- The mid-block study sections have not yet been selected, so site-specific flight planning and ground control cannot be finalised
- A detection and multi-object tracking pipeline has been implemented and tested on a sample video to validate the workflow
- A second pretrained model, run on a sample video, could not identify motorcycles, showing that such models are not reliable for local traffic
- Trajectories have not been generated in real-world coordinates, because ground coordinates are not yet available

6.3 PIPELINE TESTING ON THE SAMPLE VIDEO

Before field deployment, the detection and tracking pipeline was tested on a sample UAV video to validate the overall workflow. About 98 representative frames were extracted for visual inspection only, which is distinct from the video data used for detection and tracking: the complete video was used for detection and tracking. The details are given in Table 6.1.

Table 6.1: Details of the sample video and processing environment

Item	Detail
Video resolution	Approximately 1280 × 736 pixels
Frame rate	Approximately 24–25 FPS
Total frames	Approximately 477
Frames extracted for visual inspection	About 98 representative frames (inspection only)
Computing environment	Google Colab with NVIDIA Tesla T4 GPU
Software	Python, PyTorch, OpenCV, Ultralytics YOLO and ByteTrack
Data storage	Video stored on Google Drive

6.4 VEHICLE DETECTION RESULTS

Detection used a pretrained DroneFlow / VisDrone vehicle-detection model based on YOLOv8, loaded through the Ultralytics framework. The confidence threshold was set to 0.35 and the inference image size to 960 pixels. The vehicle classes observed in the output were cars, vans, trucks, motorcycles and buses. For every frame the output consists of the bounding boxes, class labels and confidence scores of the detected vehicles. Figure 6.1 shows one frame of the output, with the detections and the tracking IDs described in the next section.

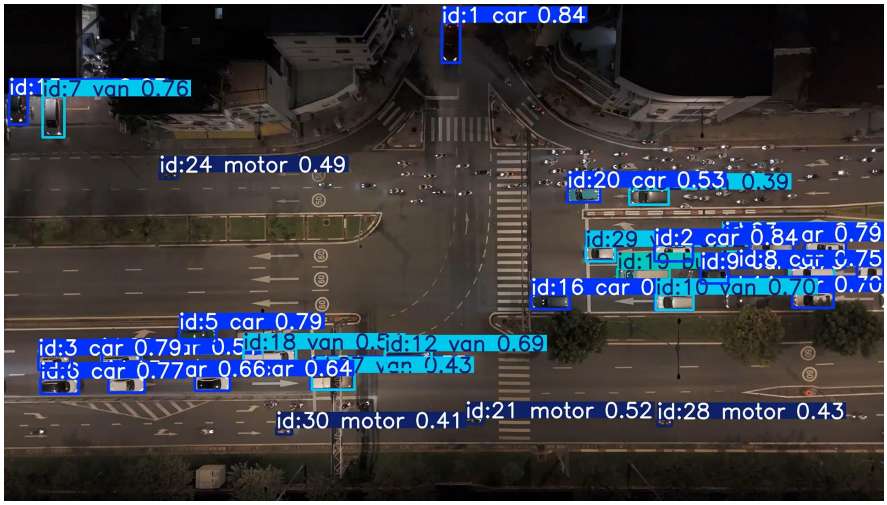


Figure 6.1: Sample frame showing detection and tracking output (class, confidence and track ID)

6.5 MULTI-OBJECT TRACKING RESULTS

Tracking used ByteTrack (bytetrack.yaml) to maintain the identity of each vehicle across frames. The workflow was: vehicle detection, bounding boxes, ByteTrack, persistent vehicle IDs and vehicle movement paths. About 470 of the 477 frames were successfully processed during tracking, with GPU acceleration on the Tesla T4. The output video shows the bounding boxes, class labels, confidence values and persistent tracking IDs for frame-by-frame vehicle movement, as in Figure 6.1.

6.6 TRACKING OUTPUT IN THE PIXEL DOMAIN

The tracker output was collected into a table with one record per vehicle per frame. The table holds 19,133 records and 614 unique vehicles (track IDs). Its columns are listed in Table 6.2, and the first five records are shown in Table 6.3.

Table 6.2: Structure of the tracking output table

Column	Meaning
frame	Frame number in the video
time_sec	Time of the frame in seconds
track_id	Persistent identity assigned by ByteTrack
class_id	Class label index from the detector
confidence	Detection confidence score
cx, cy	Anchor point of the box in image pixels (cx is the box centre in x; cy equals the bottom edge y2)
x1, y1, x2, y2	Top-left and bottom-right corners of the bounding box in image pixels

Table 6.3: First five records of the tracking output (pixel coordinates)

frame	time_sec	track_id	class_id	confidence	cx	cy	x1	y1	x2	y2
1	0.033367	1	0	0.836275	909.82	1681.20	831.41	1615.15	988.24	1681.20
1	0.033367	2	0	0.833226	535.16	1689.86	455.08	1617.11	615.24	1689.86
1	0.033367	3	0	0.824054	223.97	1595.78	152.48	1526.33	295.45	1595.78
1	0.033367	4	0	0.816103	244.28	1695.93	159.00	1623.77	329.56	1695.93
1	0.033367	5	0	0.814664	3071.04	1215.17	2982.47	1150.58	3159.60	1215.17

The first record corresponds to track 1 of Figure 6.1, a car detected with a confidence of 0.84. The positions are image coordinates in pixels and carry no metric meaning, so they cannot be reported as distance or speed until they are converted to ground coordinates (Section 6.9).

6.7 RELIABILITY CHECK OF A SECOND PRETRAINED MODEL

A second pretrained vehicle-detection model was run on a sample video to check its reliability on local mixed traffic. It failed to identify motorcycles and other two-wheelers, which form a dominant class in Indian traffic. In the output crop in Figure 6.2, labels appear only on the larger vehicles, whereas the small two-wheelers visible in the scene are left without boxes.



Figure 6.2: Output of the second pretrained model on a sample video (two-wheelers not identified)

6.8 LIMITATIONS OF EXISTING METHODS

The sample-video tests show that pre-existing methods, applied as they are, are not accurate enough for mixed Indian traffic:

- A second pretrained model could not identify motorcycles and other two-wheelers
- In the output of the first model (Figure 6.1), the motorcycle detections carry confidence scores of about 0.41–0.52, which are generally lower than those of cars (up to 0.84), consistent with the difficulty of detecting small objects
- The classes observed (cars, vans, trucks, motorcycles and buses) do not follow the five-class scheme of this study, and three-wheelers are not among them
- Tracking worked only in the pixel domain; without ground coordinates the results have no physical scale

These observations are visual. Quantitative accuracy will be reported only after the local dataset has been annotated. They nevertheless support the need for a small-object-optimised

detector trained on a local five-class dataset, which is the approach proposed in the methodology, and agree with the benchmark evidence that generic detectors lose accuracy under UAV conditions (Du et al., 2018).

6.9 TRAJECTORY GENERATION: CURRENT STATUS

Vehicle detection and tracking in the pixel domain have been completed on the sample video. **Trajectories in real-world coordinates have not been found**: converting the tracked IDs into ground coordinates needs DGPS ground control, which is not available because the coordinates have not been given. Until the geometric transformation of Section 5.11 is carried out, pixel positions give only relative movement and cannot be reported as distance, speed, TTC or PET.

6.10 DISCUSSION

The sample-video work shows that the detection and tracking part of the workflow can be run end to end: vehicles are detected, given persistent IDs and collected into a trajectory table. It also shows where the workflow cannot yet go. The reliability check confirms the main premise of the study, that an off-the-shelf detector cannot be trusted for two-wheeler-heavy traffic, so the baseline and improved detectors must be trained on local data. The missing step is georeferencing, which depends on the purchase of the UAV, the selection of the sections and the DGPS survey. These field activities are therefore the critical path of the project, as reflected in the schedule in Chapter 7.

CHAPTER-7
CONCLUSIONS AND WORK PLAN

7.1 GENERAL

This interim report presents the progress of the project on UAV-based vehicle detection and trajectory analysis for improving highway traffic safety. It summarises the work completed, the results from the sample video, the expected outcomes and the plan for the remaining work.

7.2 SUMMARY OF WORK DONE

- A review of 17 studies on UAV traffic monitoring, vehicle detection, tracking, georeferencing and surrogate safety measures was completed, and the research gaps were identified
- The objectives, data requirements and a detailed methodology covering twelve steps, with the governing equations, were defined
- A detection and tracking pipeline was tested on a sample UAV video, producing a pixel-coordinate tracking table of 19,133 records for 614 tracked vehicles
- A second pretrained model failed to identify motorcycles, which shows that existing pretrained models are not reliable for local two-wheeler-heavy traffic
- Trajectories in real-world coordinates have not been found, because ground coordinates are not yet available

7.3 EXPECTED OUTCOMES

- A DGPS-georeferenced, annotated UAV vehicle dataset representative of Thiruvananthapuram traffic conditions
- A small-object-optimised YOLO model with measurable accuracy gains over baseline YOLO for high-altitude vehicle detection
- Validated vehicle count, speed and traffic density estimates for the surveyed mid-block sections
- A crash prediction model derived from UAV-based trajectory data and surrogate safety measures
- Identification of accident hotspots and level-of-service assessment at the study locations, presented through visualisation of the derived traffic safety metrics

7.4 CHALLENGES AND NEXT STEPS

- Select and finalise the mid-block study sections in Thiruvananthapuram (not yet selected), using the proposed criteria
- Purchase the UAV (DJI Neo 2) and carry out the field survey with DGPS ground control
- Annotate vehicle classes in CVAT on imagery from the finalised locations
- Train and evaluate the baseline and improved small-object YOLO detectors on the local dataset
- Generate trajectories from the tracking output as time-ordered coordinate sequences
- Apply DGPS georeferencing, then compute TTC, PET and the crash prediction model

7.5 SCHEDULE OF WORK

The plan for the remaining work is given in Table 7.1. “Done” marks work carried out till date and “Plan” marks work still to be done, by month.

Table 7.1: Schedule of work

Activity	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Topic Finalisation	Done								
Literature Review	Done	Done	Plan	Plan					
UAV Data Acquisition			Plan	Plan					
Preprocessing		Done	Done	Plan	Plan				
CVAT Annotation		Done			Plan	Plan			
Model Training		Done				Plan	Plan		
Tracking & Trajectory							Plan	Plan	
Safety & LOS Analysis								Plan	Plan
Report Writing & Review									Plan

7.6 CONCLUSION

The study aims to show that UAV imagery, a small-object-optimised detector, multi-object tracking and DGPS georeferencing can be combined into a reproducible workflow that gives vehicle counts, speeds, densities, surrogate safety measures and crash risk for mid-block

highway sections in Thiruvananthapuram. The work so far confirms the need for such a workflow: pretrained detectors were found unreliable for two-wheelers, and pixel-domain tracking alone cannot give physical measures.

The remaining work depends on the field activities of selecting the sections, purchasing the UAV and carrying out the DGPS survey. Once these are done, the dataset can be annotated, the detectors trained and compared, and the trajectories georeferenced and analysed for safety.

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